

Enhancing UAV Swarm Resilience to Payload Failures: A Bidirectional Task Reallocation Approach

Yifan Zeng[✉], Linghua Wu, Jinning Li[✉], and Xuebin Zhuang[✉]

Abstract—The resilience of uncrewed aerial vehicle (UAV) swarms is critical for ensuring system performance in challenging environments. Existing resilience research predominantly employs binary failure models, which overlook the residual capabilities of UAVs following partial payload failure and consequently limit the effectiveness of recovery strategies. To address this gap, this article proposes a payload-level swarm resilience enhancement framework. First, a payload-oriented state model is constructed, correlating swarm performance with the functional status of payloads. Second, a bidirectional task reallocation (BDTR) method is introduced as a proactive enhancement mechanism. This method adopts a dynamic “task offloading followed by task acquisition” interaction to fully leverage the swarm’s residual capabilities and achieve load balancing. Simulation results demonstrate that BDTR significantly outperforms benchmark methods in comprehensive resilience while maintaining the highest mission completion rate. Validation through Monte Carlo simulations shows an average mission completion rate of 0.829 and an average resilience index of 0.862, mitigating the tradeoff of “high completion rates with poor execution efficiency” observed in conventional baseline methods. This study offers a new theoretical foundation and practical pathway for enhancing UAV swarm resilience, supporting the development of more reliable autonomous systems.

Index Terms—Bidirectional task reallocation (BDTR), payload failures, resilience, uncrewed aerial vehicle (UAV) swarm.

I. INTRODUCTION

UNCREWED aerial vehicle (UAV) swarms function as pivotal assets in complex missions, including cooperative search [1], [2], emergency response [3], [4], and dynamic tasking [5], [6], owing to their distinct coordination and flexibility advantages [7], [8], [9]. However, susceptibility to payload failures [10], communication constraints [11], and environmental uncertainties [12], [13] persists.

In addressing these failures, *high-precision state perception* constitutes the foundation for resilient decision-making. Although the multi-state system (MSS) theory [14], [15] establishes that system performance degrades gradually, current research predominantly adopts a “*platform-centric*” *binary model* [16]. Consequently, decision models often discard partially damaged UAVs where a single sensor failure renders the

entire platform unavailable. This lack of *functional granularity* fails to capture the complex health status defined by multidimensional functional states, leading to the premature loss of nodes with *residual capabilities*.

To address the limitations of existing binary models and insufficient utilization of residual capabilities, this article proposes a payload-level resilience framework. The main contributions are as follows.

1) *Established a payload-oriented state model*: This framework expands the UAV state representation from a scalar binary value to a *Payload-level Health Vector*, achieving precise, real-time capture of residual heterogeneous capabilities.

2) *Proposed a dynamic resilience assessment model*: Integrating task throughput, payload utilization, and timeliness, this model quantifies recovery quality at the micro-level, providing a comprehensive standard for heterogeneous swarms.

3) *Designed a Bidirectional Task Reallocation (BDTR) algorithm*: Leveraging fine-grained state information, BDTR employs a “task offloading and acquisition” mechanism to maximize residual capability utilization, significantly outperforming unidirectional baselines.

The rest of this article is organized as follows. Section II reviews related work. Section III details the payload-oriented state model. Section IV presents the resilience metric and the BDTR algorithm. Section V analyzes simulation results. Finally, Section VI concludes this article.

II. RELATED WORKS

Research on UAV swarm task allocation generally falls into three categories: market-based approaches, meta-heuristics, and fast heuristics.

Market-based approaches, including the Contract Net Protocol [17] and consensus based bundle algorithm (CBBA) [18], support decentralization. Distributed Task Reallocation serves as the core mechanism for swarms to maintain performance in dynamic environments. A representative approach in this domain is the Swarm-GAP algorithm [19], which establishes a decentralized framework where UAVs negotiate assignments via token passing based on their specific capabilities and task preferences. Recent advances involving consensus-based auctions [6], [20] have further improved coordination. However, these methods incur significant communication overhead during topology fragmentation, thereby increasing the difficulty of real-time processing and the uncertainty of task reconstruction.

Received 18 October 2025; revised 18 January 2026; accepted 26 February 2026. Date of publication 17 March 2026; date of current version 22 April 2026. This work was supported by the Science and Technology on Information System Engineering Laboratory under Grant 05202007. (Corresponding author: Xuebin Zhuang.)

The authors are with the School of Systems Science and Engineering, Sun Yat-sen University, Guangzhou 510006, China (e-mail: zhuangxb@mail.sysu.edu.cn).

Digital Object Identifier 10.1109/JSYST.2026.3669721

TABLE I
COMPARATIVE ANALYSIS OF BDTR AND OTHER SOTA CATEGORIES REGARDING CHARACTERISTICS AND APPLICABILITY

Algorithm Category	Typical Algorithms	Time Complexity	Global Optimality	Real-time Responsiveness	Applicability to Resilience	Limitations in This Context
Meta-heuristics	GA, PSO, ACO	High Exponential	High Theoretically approaches global optimum	Low Slow iteration convergence; unable to meet millisecond-level dynamic re-planning requirements	Not Applicable Suitable for distributed scenarios with good communication, but consensus convergence is difficult when node failures cause network topology fragmentation.	Under high dynamic interference, the battlefield environment may have already changed by the time GA calculation is completed (timeliness lag).
Market-based	CBBA, swarm-GAP	Medium Polynomial	Medium Typically converges to 50% optimality	Medium Limited by communication bandwidth and consensus convergence rounds	Partially Applicable Suitable for distributed scenarios with good communication, but consensus convergence is difficult when node failures cause network topology fragmentation.	In failure scenarios with restricted communication or unstable links, negotiation mechanisms are prone to deadlocks or timeouts.
Fast Heuristics	SROM, RRAM	Low Linear / Low Polynomial	Low Sacrifices quality for speed	High Instantaneous response	Applicable Current mainstream industrial solution, but often yields “feasible solutions” rather than “quality solutions”.	Prone to becoming trapped in local minima, resulting in low mission completion rates and unbalanced resource utilization.
Bidirectional Task Reallocation (Proposed)	BDTR	Low Low Polynomial	Sub-optimal Significantly outperforms SROM, approaching GA	High Converges with only a few iterations	Optimal Achieves the best trade-off between solution quality and computational efficiency.	Does not guarantee global optimality, but is the optimal choice for time-sensitive resilience recovery tasks.

Metaheuristics, encompassing genetic algorithms (GAs) [21], particle swarm optimization (PSO) [22], and ant colony optimization (ACO) [23], offer global optimality but suffer from exponential time complexity, rendering them ill-suited for millisecond-level dynamic recovery.

Regarding resilience metrics, while some studies focus on network topology against cascading failures [24], [25] or operational kill chains [26], [27], quantifying mission-level recovery remains critical. Consequently, in scenarios where recovering performance [28], [29] is paramount, fast heuristics are often preferred for their responsiveness. Two baseline strategies are prominent in this category. The *Random Resource Allocation Method (RRAM)* [30] serves as a zero-information baseline. It focuses solely on current task completion but lacks sustainability checks for future tasks. The *Soft Resource Optimization Method (SROM)* [31], [32] employs resilience-oriented dynamic reallocation, balancing current completion with future resilience. However, SROM utilizes a unidirectional offloading mechanism, ignoring the residual capabilities of damaged nodes.

Critically, most existing works [22], [33], including the aforementioned baselines, predominantly employ “platform-centric” binary failure models. In contrast, the proposed BDTR method addresses this gap by introducing payload-level granularity. It balances polynomial efficiency with the effective exploitation of residual capabilities.

To clarify the positioning of different algorithm categories, a theoretical comparison is summarized in Table I. Despite extensive optimization, most existing distributed strategies lack a “functional-level decoupling and swapping” mechanism, resulting in healthy nodes being overloaded while damaged nodes with residual capabilities are underutilized.

III. PRELIMINARIES

A. Payload-Oriented State Model for UAV Swarms

This study constructs a modeling framework for UAV swarms under payload failures, based on set theory and complex network

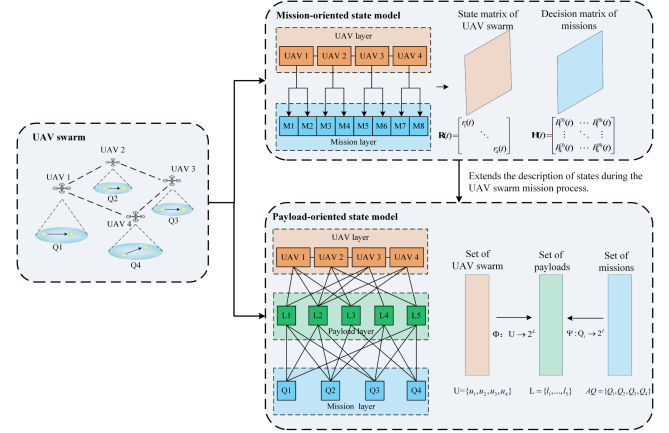


Fig. 1. Illustration of an extension framework from mission-oriented state model to payload-oriented state model for UAV swarms.

theory. Based on the mission-oriented state model [11], [34], the extended framework refines the traditional platform-centric perspective, as shown in Fig. 1. Let $U = \{u_1, \dots, u_{N_U}\}$ be the UAV set and $L = \{l_1, \dots, l_{N_L}\}$ be the payload types. The static payload configuration is $\Phi : U \rightarrow 2^L$. To describe dynamic reliability, the *binary health status* is defined by $\sigma : U \times L \rightarrow \{0, 1\}$. Consequently, the *effective payload subset* is defined as $\Phi_{\text{eff}}(u_i) = \{l_k \in \Phi(u_i) \mid \sigma(u_i, l_k) = 1\}$.

Regarding mission execution, let AQ be the task set and $Q_i = \{q_{i1}, \dots, q_{i, N_{Q_i}}\}$ be the task queue of u_i . The task dependency is mapping $\Psi : Q_i \rightarrow 2^L$. Finally, the task execution capability is determined by matching resources: $TC(u_i, q_{ij}) = 1$ if $\Psi(q_{ij}) \subseteq \Phi_{\text{eff}}(u_i)$, and 0 otherwise. This defines the capable set $U_{\text{cap}}(q_{ij}) = \{u_i \in U \mid TC(u_i, q_{ij}) = 1\}$.

B. Mission Throughput Model of UAV Swarm

This study extends the throughput concept [35] to the mission level. As illustrated in Fig. 2, each UAV processes task set Q_i . The directed arrows signify the execution flow governed

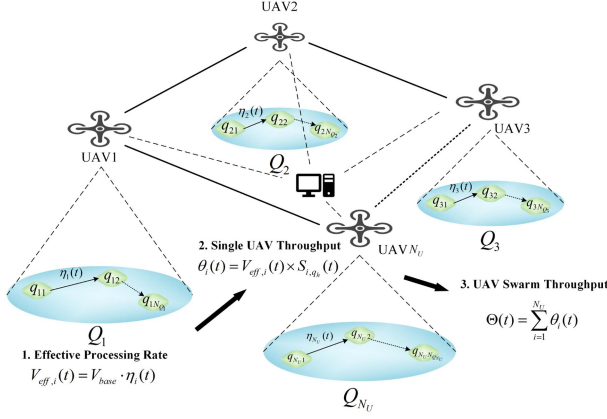


Fig. 2. Mission-oriented UAV swarm throughput framework.

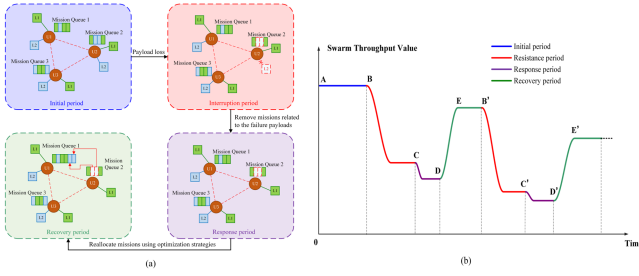


Fig. 3. Illustration of the resilience process: (a) Four sequential phases; (b) Performance response curve.

by a performance discount factor $\eta_i(t)$, allowing the system to monitor individual and swarm throughput.

The effective task processing rate for UAV i is

$$V_{eff,i}(t) = V_{base} \cdot \eta_i(t) \quad (1)$$

where $V_{base} = \min_{k \in U} (1/|Q_k(0)|)$ is the baseline rate.

The performance discount factor $\eta_i(t)$ accounts for overload degradation

$$\eta_i(t) = 1 \text{ if } |Q_i(t)| \leq N_{Mi}, \text{ else } \Delta \quad (2)$$

where N_{Mi} is the full-performance threshold and Δ is the degradation rate. The task processing status $S_{i,qh}(t)$ indicates feasibility

$$S_{i,qh}(t) = 1 \text{ if } q_h \in Q_i(t), \text{ else } 0. \quad (3)$$

The individual task throughput $\theta_i(t)$ and total swarm throughput $\Theta(t)$ are defined as

$$\theta_i(t) = V_{eff,i}(t) \times S_{i,qh}(t), \Theta(t) = \sum_{i=1}^N \theta_i(t). \quad (4)$$

C. Resilience Process of UAV Swarm

Fig. 3 illustrates the resilience process comprising four phases. In the *initial phase*, the swarm operates in a baseline state. The *disruption phase* begins with a payload failure, severing collaborative links. The *response phase* is a transitional

period for strategy planning. Finally, in the *recovery phase*, optimized reallocation drives throughput rebound. Fig. 3(b) depicts this via performance curves: from baseline (A) to degradation (C), followed by a plateau (D), and finally converging to a recovered steady state (E).

IV. RESILIENCE ASSESSMENT AND ENHANCEMENT METHOD FOR MISSION-PAYLOADS

To address swarm performance degradation caused by payload failures, this section proposes a closed-loop resilience enhancement framework. First, a comprehensive resilience metric is defined; second, a mathematical optimization model is constructed; finally, the BDTR algorithm is proposed.

A. Comprehensive Resilience Metrics

For UAV swarms, resilience assessment must balance “quantity” and “speed.” Based on the frameworks by Tran et al. [29] and Cheng et al. [34], we construct a 2-D assessment system.

First, the *Task Resilience Factor* (R_{task}) quantifies the restoration of execution capability

$$R_{task} = \frac{\sum_{i=1}^{N_U} \sum_{q_{ij} \in Q_i} \Omega(q_{ij}) + \gamma \cdot N_{realloc}}{\lambda_{max} \cdot (\tilde{t}_{end} - \tilde{t}_{start})} \quad (5)$$

where $\Omega(q_{ij}) \in \{0, 1\}$ indicates effective completion; $\gamma \in [0, 1]$ is the reallocation gain coefficient; and $N_{realloc}$ is the number of salvaged tasks.

Second, the *Time Resilience Factor* (R_{time}) penalizes the extra time overhead to reflect recovery agility

$$R_{time} = 1 - \frac{T_{recover}}{\tilde{t}_{end} - \tilde{t}_{start}}. \quad (6)$$

Here, $T_{recover}$ denotes the recovery duration from performance drop to steady-state equilibrium.

Finally, the *Cumulative Resilience Index (CRI)* synthesizes these aspects

$$R_{total} = \alpha \cdot R_{task} + (1 - \alpha) \cdot R_{time} \quad (7)$$

where α is the weight coefficient. Maximizing R_{total} implies completing maximum high-value tasks within minimal time.

B. Mathematical Modeling of the Reallocation Problem

The article formalizes this problem as a dynamic generalized assignment problem (GAP). Define binary variable $H(u_i, q_{ij}) \in \{0, 1\}$. Maximizing resilience R_{total} is mathematically equivalent to minimizing the system’s maximum load while ensuring completed tasks

$$\max_H R_{total}(H) \quad (8)$$

$$\text{s.t.} \quad \sum_{u_i \in U_{cap}(q_{ij})} H(u_i, q_{ij}) = 1, \quad \forall q_{ij} \in \mathcal{K}_{active} \quad (C1)$$

$$\sum_{q_{ij} \in Q_i} H(u_i, q_{ij}) \leq C(u_i), \quad \forall u_i \in U \quad (C2)$$

$$T_{fly} + T_{exec} + \tau_{comm} \leq T_{dead} \quad (C3)$$

$$\Psi(q_{ij}) \subseteq \Phi_{\text{eff}}(u_i), \quad \text{if } H(u_i, q_{ij}) = 1. \quad (\text{C4})$$

1) *Constraint Explanation and Physical Interpretation*: To ground the model in physical realities

(C1) *Uniqueness*: Ensures each active task is assigned to exactly one capable UAV. $\mathcal{K}_{\text{active}}$ represents the set of pending tasks.

(C2) *Capacity*: The tasks must not exceed the limit $C(u_i)$. Physically, this maps to finite onboard resources (energy/compute), preventing single-node exhaustion and cascading failures.

(C3) *Timeliness*: $T_{\text{fly}} + T_{\text{exec}} + \tau_{\text{comm}} \leq T_{\text{dead}}$. This explicitly accounts for flight maneuverability (T_{fly}) and network latency (τ_{comm}) in bandwidth-limited environments, ensuring feasibility under non-ideal communication. T_{dead} is the maximum deadline of a single task.

(C4) *Capability*: The UAV must possess the specific functional payloads required by the task. This is the core constraint in payload failure scenarios.

C. Bidirectional Task Reallocation Algorithm

BDTR decomposes the NP-hard GAP into two serialized operators: 1) Feasibility Restoration and 2) Residual Capacity Exploitation.

1) Theoretical Basis of the Algorithm

Phase 1: Feasibility Restoration Operator When u_{fail} fails, its effective payload set shrinks to Φ'_{eff} . Invalid tasks $\mathcal{K}_{\text{invalid}} = \{q \in Q_{\text{fail}} \mid \Psi(q) \not\subseteq \Phi'_{\text{eff}}(u_{\text{fail}})\}$ are identified. To restore feasibility minimizing makespan growth, the algorithm reallocates each invalid task χ to the capable candidate with minimum load

$$u_{\text{target}}^* = \arg \min_{u_j \in U_{\text{cand}}(\chi)} |Q_j| \quad (9)$$

where $U_{\text{cand}}(\chi)$ is the feasible candidate set. This guarantees minimum growth of the load vector's L_∞ norm.

Phase 2: Residual Capacity Exploitation Operator This operator utilizes the residual capability of u_{fail} via “task swapping.” Let $\mathcal{U}_{\text{donor}}$ be the max-load node set. For donor u_d , the swappable set is

$$\mathcal{K}_{\text{swap}} = \{q \in Q_d \mid \Psi(q) \subseteq \Phi'_{\text{eff}}(u_{\text{fail}})\}. \quad (10)$$

A swap executes if transfer gain $\Delta G = |Q_d| - |Q_{\text{fail}}| > 1$, seeking the descent gradient on the heterogeneous constraint manifold.

2) *Convergence Analysis*: The system state sequence $\{S_k\}$ generated by the BDTR algorithm during the residual capacity exploitation phase strictly decreases the system load variance and converges to a locally optimal state in finite steps.

Proof: We construct a Lyapunov potential function $\mathcal{P}(S)$ to characterize the load imbalance of the swarm

$$\mathcal{P}(S) = \sum_{u_i \in U} |Q_i|^2 \quad (11)$$

Consider the k th iteration where a task is transferred from a donor node u_d (with load L_d) to a recipient node u_r (with load L_r). The algorithm's triggering condition dictates that a swap occurs if and only if the load difference satisfies $L_d - L_r \geq 2$.

Algorithm 1: Bidirectional Task Reallocation Algorithm (BDTR).

Require: UAV set U ; Task queues $\{Q_i\}$; Failed node u_{fail} .

Ensure: Optimized task queues $\{Q'_i\}$.

Phase 1: Feasibility Restoration

1: Identify invalid tasks $\mathcal{K}_{\text{invalid}}$ and **Sort** them in descending order.

2: Update queue: $Q_{\text{fail}} \leftarrow Q_{\text{fail}} \setminus \mathcal{K}_{\text{invalid}}$.

3: **for** each task $k \in \mathcal{K}_{\text{invalid}}$

4: Find candidates: $U_{\text{cand}} \leftarrow \{u \in U \mid \Psi(k) \subseteq \Phi_{\text{eff}}(u)\}$.

5: **if** $U_{\text{cand}} \neq \emptyset$

6: Select target: $u_{\text{target}} \leftarrow \arg \min_{u \in U_{\text{cand}}} |Q_u|$.

7: Assign: $Q_{\text{target}} \leftarrow Q_{\text{target}} \cup \{k\}$.

8: **end if**

9: **end for**

Phase 2: Residual Capacity Exploitation

10: **while** True

11: Select donor $u_d \in \mathcal{U}_{\text{donor}}$ with max load.

12: **if** $|Q_d| - |Q_{\text{fail}}| \leq 1$ **break**

13: **end if**

14: Find swap tasks

$\mathcal{K}_{\text{swap}} \leftarrow \{q \in Q_d \mid \Psi(q) \subseteq \Phi'_{\text{eff}}(u_{\text{fail}})\}$.

15: **if** $\mathcal{K}_{\text{swap}} \neq \emptyset$

16: Transfer optimal task k^* : $Q_d \rightarrow Q_{\text{fail}}$.

17: **else**

18: Remove u_d from $\mathcal{U}_{\text{donor}}$ and continue.

19: **end if**

20: **end while**

21: **return** $\{Q'_i\}$.

Let the system state transition from S_k to S_{k+1} . The loads update as $L'_d = L_d - 1$ and $L'_r = L_r + 1$. The change in the potential function, denoted as $\Delta \mathcal{P}$, is derived as follows:

$$\begin{aligned} \Delta \mathcal{P} &= \mathcal{P}(S_{k+1}) - \mathcal{P}(S_k) \\ &= [(L_d - 1)^2 + (L_r + 1)^2] - [L_d^2 + L_r^2] \\ &= -2(L_d - L_r - 1). \end{aligned} \quad (12)$$

Applying the triggering condition $L_d - L_r \geq 2$, we obtain

$$\Delta \mathcal{P} \leq -2(2 - 1) = -2 < 0. \quad (13)$$

This inequality implies that every valid task swapping operation strictly reduces the system potential function $\mathcal{P}(S)$. Since the configuration space of task assignments is finite and $\mathcal{P}(S)$ is bounded below by zero, the sequence $\{\mathcal{P}(S_k)\}$ must converge, guaranteeing the termination of the algorithm. $\square$

3) *Algorithm Implementation and Complexity*: The BDTR process is visually illustrated in Fig. 4, and the detailed execution logic is presented in Algorithm 1.

Following the algorithmic structure, the computational complexity is primarily dominated by the sorting operation ($O(N_{\text{fail}} \log N_{\text{fail}})$) and the candidate matching process in Phase 1 ($O(N_{\text{fail}} \cdot N_U \cdot N_L)$), alongside the linear donor search in Phase 2. Consequently, the aggregate time complexity is derived as approximately $O(N_Q \cdot N_U \cdot N_L)$.

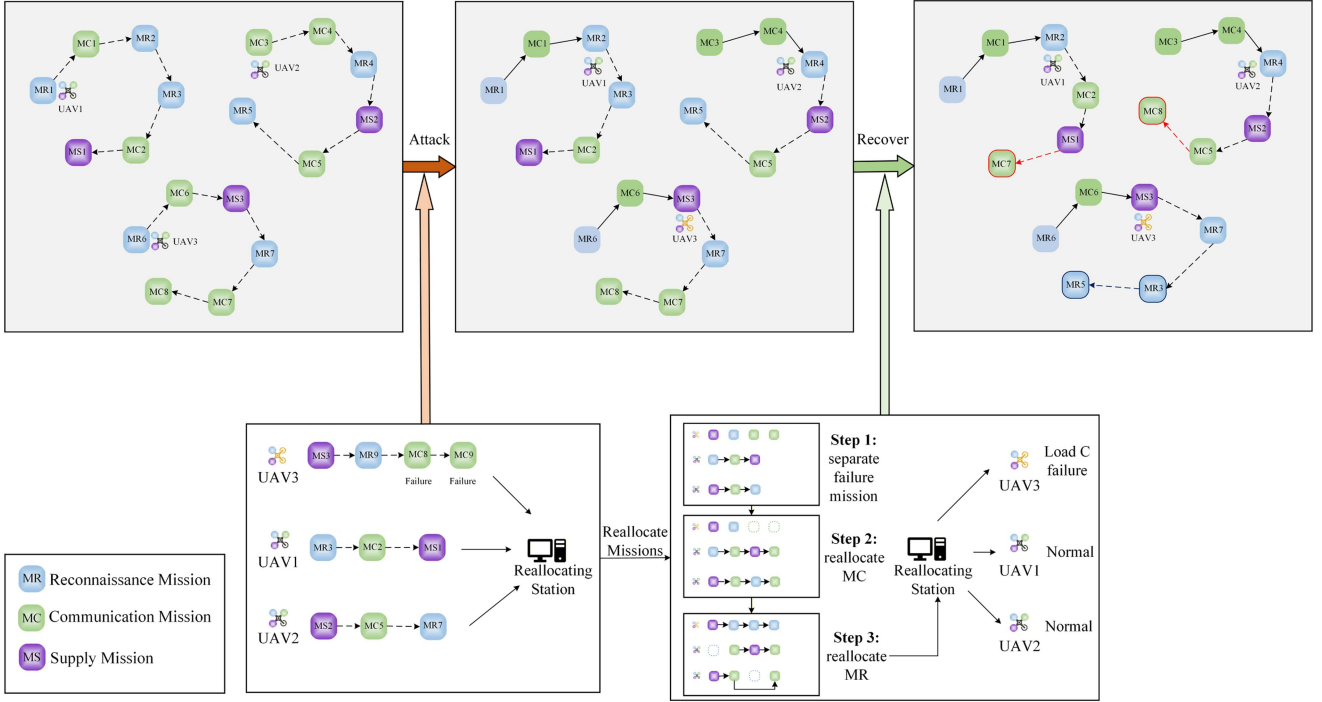


Fig. 4. Illustration of the Bidirectional Task Reallocation (BDTR) process.

Notably, regarding the scalability with payload diversity (N_L), the proposed method offers robust performance from two perspectives. *Theoretically*, BDTR maintains *linear scalability* with N_L , exhibiting a significant advantage over exact combinatorial algorithms that typically suffer from exponential complexity growth ($O(k^{N_L})$). *Practically*, while current engineering swarms typically operate with a finite set of functional payload classes (e.g., Reconnaissance, Communication, and Attack, where $N_L \leq 10$), this linear property guarantees that the algorithm remains computationally efficient even in future complex missions involving highly heterogeneous payload configurations.

V. NUMERICAL EXPERIMENTS

A. Data Preparation and Experiment Setup

The simulation experiments were conducted in MATLAB R2024a on an Intel i7-12700H PC with 16 GB RAM. To evaluate scalability and robustness, two distinct scenarios (Small-scale versus Large-scale) were established, with key parameters detailed in Table II. A total of 300 Monte Carlo simulations were performed for each scenario to ensure statistical significance.

Simplification of Physical Constraints: To strictly isolate algorithmic efficacy from hardware-specific variances, the simulation adopts a standardized execution timeline. While the theoretical model (Section IV-B) incorporates physical constraints like maneuverability (T_{fly}) and latency (τ_{comm}), these factors are disregarded in experiments to focus exclusively on the decision-making and load-balancing capability of BDTR.

In each run, the swarm faced persistent payload failures under three interruption strengths (20%, 30%, 40%) to test performance in adversarial environments.

TABLE II
SIMULATION AND SCENARIO PARAMETERS

Category	Parameter	Value / Description
General Settings	Δ (Degradation)	0.9
	α (Weight)	0.5 (R_{task} vs R_{time})
	γ (Realloc. Gain)	0.5
	N_{sim} (Monte Carlo)	300 runs
Scenario 1 (Small)	Swarm Size (N_U)	5 UAVs
	Complexity	2 Payload types, 80 Tasks
Scenario 2 (Large)	Swarm Size (N_U)	10 UAVs
	Complexity	3 Payload types, 120 Tasks

B. Mission Throughput Model Validation

A series of simulation experiments were conducted to validate the effectiveness of the proposed swarm throughput model. The analysis focuses on the performance of the bidirectional task reallocation algorithm when subjected to three distinct and systematic attack patterns. As illustrated in the Fig. 5, the results clearly demonstrate the process of dynamic degradation in swarm task throughput over time for each scenario. The degradation of the swarm's performance exhibits a distinct gradient that corresponds to the severity of the attack pattern. The system demonstrated the highest resilience under the Random Attack scenario, followed by the Balance Attack, while the Directed Attack resulted in the most rapid performance degradation. This hierarchical outcome aligns perfectly with theoretical expectations.

In the Random Attack scenario, the system exhibits the most gradual performance degradation because the damage is distributed across all payload types. This distributed attack pattern

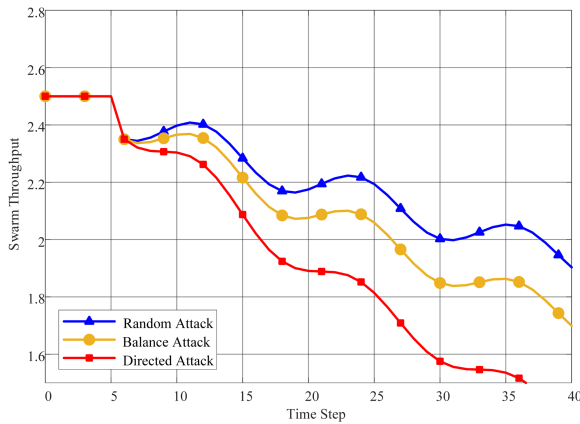


Fig. 5. Swarm throughput trajectories under different attack modes.

preserves maximum system flexibility and the largest number of potential exchange partners for the bidirectional reallocation algorithm. This enables the algorithm to mitigate the impact of individual UAV failures most effectively through continuous and efficient load balancing. The model's output, a slowly declining curve, accurately reflects this high degree of resilience. In contrast, the Directed Attack scenario proved to be the most destructive. By concentrating its attack on a single payload type, this strategy systematically eliminates a specific mission capability from the swarm. This rapidly depletes the pool of UAVs capable of performing a particular task, leading to an increased rate of exchange failures and a cascading backlog of unserviceable tasks. The model accurately captures this systemic collapse—which occurs when a critical swarm capability is effectively “decapitated”—by generating a curve with a steep, nonlinear downward trend. The performance under the Balance Attack scenario logically falls between these two extremes. This attack pattern is more focused than a purely random attack but less destructive than a specialized directed attack. The swarm's flexibility is impaired but not entirely eliminated, resulting in a moderate rate of performance degradation. The model correctly positions this performance curve between the two aforementioned extremes. In conclusion, a strong correlation exists between the design of the attack patterns and the performance curves generated by the throughput model. This correlation provides robust evidence for the model's validity. It demonstrates that the model is capable not only of tracking the process of performance degradation but also of reflecting how the swarm's mission execution capabilities are compromised under different modes of attack. This validates its utility as an analytical tool for assessing swarm resilience.

C. Validation of the BDTR Resilience Optimization Method

To quantitatively evaluate the effectiveness of the proposed BDTR reallocation strategy, this study analyzes its performance under three different attack modes. Fig. 6(a) shows the cumulative resilience index (CRI) of the swarm after each stage of the three attacks, while Fig. 6(b) presents the event-specific Throughput Recovery values corresponding to each attack.

The research findings provide strong evidence for the effectiveness of the strategy and reveal its performance characteristics under different attack modes. As shown in Fig. 6(a), a performance hierarchy is maintained across all stages of the mission. The swarm exhibits the highest resilience under random attacks (CRI values of 0.926, 0.850, and 0.784, respectively), followed by balanced attacks (0.912, 0.808, and 0.718), and the lowest resilience under targeted attacks (0.868, 0.715, and 0.596). This indicates that the algorithm's effectiveness is strongly correlated with the remaining systemic flexibility within the swarm. Random attacks disperse damage across all payload types, preserving the maximum number of potential swap partners for the reallocation mechanism, thereby enabling it to effectively maintain load balance. Fig. 6(b) offers a more granular analysis of the algorithm's immediate recovery capability after each attack. The Throughput Recovery metric quantifies how much performance the BDTR algorithm salvages compared to a do-nothing baseline. A significant trend is noteworthy: The recovery values under random attacks are consistently the highest (0.487, 1.029, and 1.262), while those under targeted attacks are the lowest (0.327, 0.630, and 0.714). This directly validates the core principle of the bidirectional swap strategy: its effectiveness depends on the availability of diverse and healthy agents for task exchanges. Targeted attacks systematically eliminate the diversity for specific task types, thus undermining the algorithm's ability to recover lost throughput. Although the swarm's absolute throughput decreases over time, the absolute amount of recovered throughput actually increases in the later stages. For instance, under random attacks, the recovery value grows from 0.487 after the first attack to 1.262 after the third. This suggests that as the swarm becomes more degraded, the contribution of each successful reallocation becomes proportionally more critical. In the later stages of the mission, a single effective swap can prevent a near-total performance collapse, which underscores the growing importance and effectiveness of the BDTR strategy as system integrity diminishes.

D. Performance Comparison of Different Methods in Multi-UAV Payload Failure Scenarios

1) *Comparative Analysis of Swarm Performance:* Fig. 7 and Fig. 8 quantitatively compare the average swarm throughput under three different strategies. Across all tested scenarios and Interruption Strengths, the performance of the three strategies exhibits a clear and consistent hierarchy: the BDTR strategy performs optimally, significantly outperforming the RRAM strategy, which in turn performs better than the SROM strategy.

Fig. 7 shows the performance curves in the dual-payload mission scenario. Initial State: In the initial phase of the mission, the swarms under all strategies operate efficiently, maintaining an initial throughput of approximately 1.2. At $t = 15$ s, the swarm experiences the first attack, and the performance of the three strategies begins to diverge. The throughput of SROM drops sharply to approximately 0.7 and does not recover. A significant drop of RRAM is also observed, with the throughput falling to approximately 0.8. Different from SROM and RRAM, the

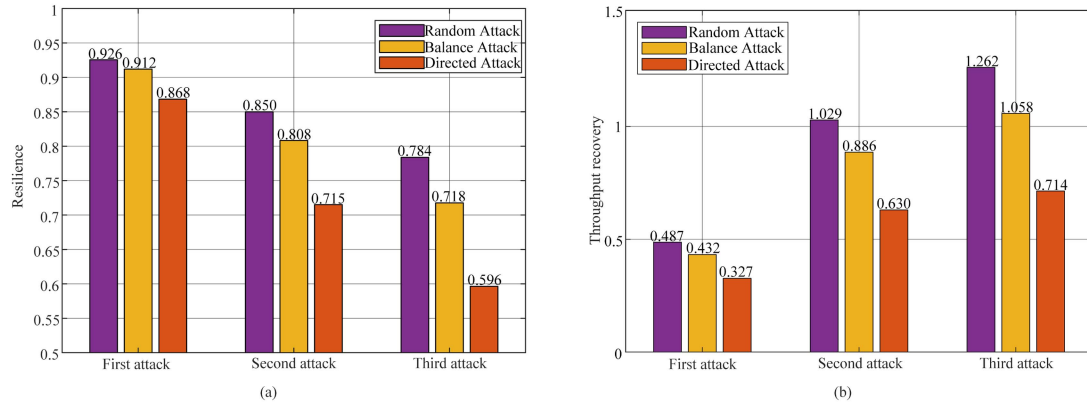


Fig. 6. Resilience and throughput improvement values of optimization strategies under different attack modes. (a) Swarm resilience comparison among three attack modes. (b) Throughput recovery comparison among three attack modes.

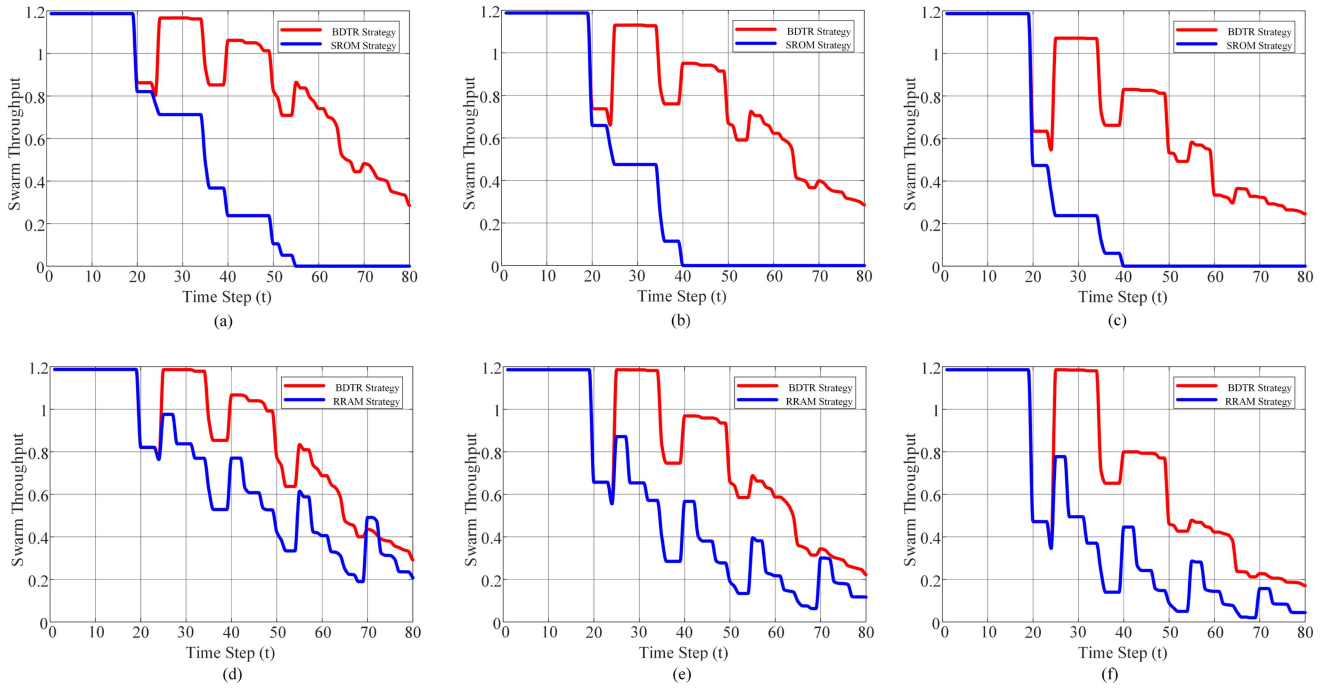


Fig. 7. Comparison of overall swarm performance with two payloads. (a)–(c) Correspond to swarm throughput comparison between BDTR and SROM strategies with interruption strengths of 20%, 30% and 40%. (d)–(f) Correspond to swarm throughput comparison between BDTR and RRAM strategies with interruption strengths of 20%, 30% and 40%.

throughput of BDTR shows only a minor decrease, quickly stabilizing at around 1.1, which demonstrates excellent immediate recovery capability. With subsequent attacks at approximately $t=25$ s, 35 s, 45 s, the performance gap between the strategies becomes more pronounced. Also, the performance curve of BDTR exhibits a smooth “graceful degradation” characteristic. Although each attack causes temporary performance fluctuations, the system consistently stabilizes at a high level. Even under the harshest 40% attack intensity [Fig. 7(f)], the swarm still maintains a throughput of approximately 0.2 at $t=80$ s, while the other two strategies have long since failed or are on the verge of failure.

Fig. 8 increases the scenario complexity to three payloads and a correspondingly larger swarm size, further testing the scalability of the algorithms. As the interruption intensity increases from 20% to 40%, the performance of all strategies declines as the environment deteriorates. However, a key phenomenon is that the performance gap between the BDTR strategy and the other strategies widens as the interruption intensity increases. This indicates that the BDTR strategy is not only effective under low-intensity disturbances, but also its performance advantage becomes even more pronounced in harsher and more challenging adversarial environments. This characteristic demonstrates that the BDTR algorithm possesses a high degree of adaptability and

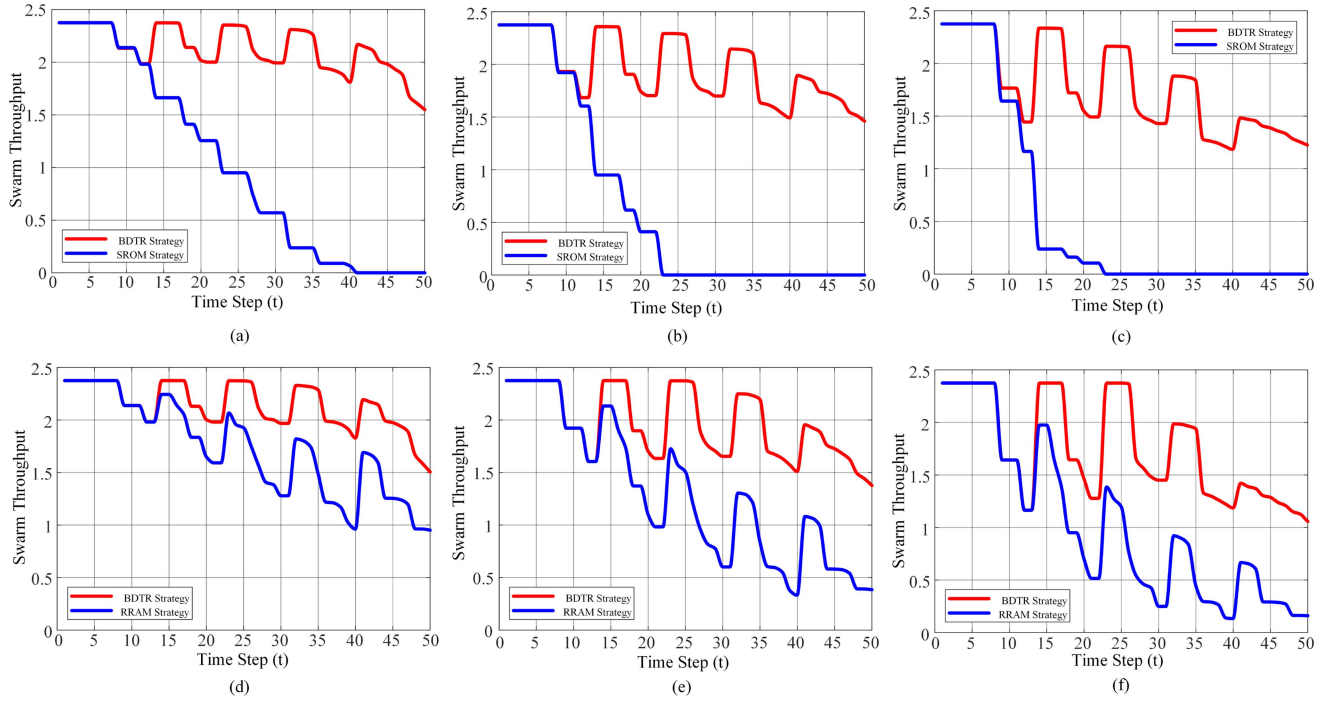


Fig. 8. Comparison of overall swarm performance with three payloads. (a)–(c) Correspond to swarm throughput comparison between BDTR and SROM strategies with interruption strength of 20%, 30% and 40%. (d)–(f) Correspond to swarm throughput comparison between BDTR and RRAM strategies with interruption strength of 20%, 30% and 40%.

excellent resilience, making it a reliable solution for maintaining mission effectiveness in dynamic and adversarial environments. In sharp contrast to the coarse-grained nature of SROM and the randomness of RRAM, BDTR’s “bidirectional swap” reallocation strategy is both fine-grained and precise. When selecting a swap partner, it rigorously verifies whether the counterpart possesses the healthy payload required to perform the relevant task. A reallocation is executed only when the swap candidate is confirmed to be fully functional for the task in question. This ensures that every reallocation is effective and can genuinely restore the system performance lost due to partial functional failures, thereby maximally maintaining the swarm’s overall mission execution capability and system resilience.

2) *Comparative Analysis of Resilience:* Figs. 9 and 10 compare the average resilience of the UAVs from an individual perspective under different payload failure frequencies.

Fig. 9 consists of five subplots, each illustrating the resilience of the five individual UAVs in the swarm as the payload failure frequency increases from 0 to 0.45 in the dual-payload scenario. For SROM strategy, the resilience of all UAVs remains 1.0 at very low failure frequencies. When the failure frequency exceeds a critical threshold, the resilience of all UAVs exhibits a cliff-edge drop. The curve has a very steep slope, indicating a sharp deterioration in system performance. As the failure frequency further increases to 0.3, the resilience value falls below 0.4. At a frequency of 0.45, the resilience has dropped below 0.2. This is a typical characteristic of “brittleness,” where the system collapses rapidly after the disturbance intensity surpasses a critical point. In contrast, for the BDTR strategy, the resilience of all UAVs remains stable at nearly 1.0 across a wide

range of failure frequencies. The curve is very flat, showing almost no impact from the disturbance intensity. After the failure frequency exceeds 0.2, the resilience begins to decline slowly and smoothly. Its degradation slope is much less steep than that of the SROM strategy, exhibiting the characteristic of “graceful degradation.” Even under the harsh condition of a failure frequency as high as 0.45, the resilience of all UAVs is maintained above 0.7. This indicates that the core functionality and efficiency of the UAVs are sustained at a very high level, even under continuous and high-intensity disturbances.

Fig. 10 extends the evaluation to a larger-scale swarm of 10 UAVs. While SROM exhibits the same brittle collapse pattern as observed in the dual-payload scenario (dropping to near 0.2), the advantage of the BDTR strategy becomes even more pronounced. The resilience curves for all 10 UAVs remain nearly horizontal at approximately 1.0 up to a failure frequency of 0.4, maintaining a high value of 0.9 even at 0.45. Crucially, comparing Figs. 10 with 9 reveals that the BDTR curves are even flatter and more stable in the 10-UAV case than in the 5-UAV case. This finding highlights the algorithm’s scalability, indicating that an increase in swarm size enhances the redundancy pool, thereby further strengthening the system’s ability to maintain resilience under BDTR.

E. Monte Carlo Simulation: A Comparison of Mission Completion Rate and Resilience Distribution

To statistically evaluate the robustness of the proposed BDTR algorithm in an uncertain environment, this study conducted 300 independent Monte Carlo simulations. The BDTR strategy

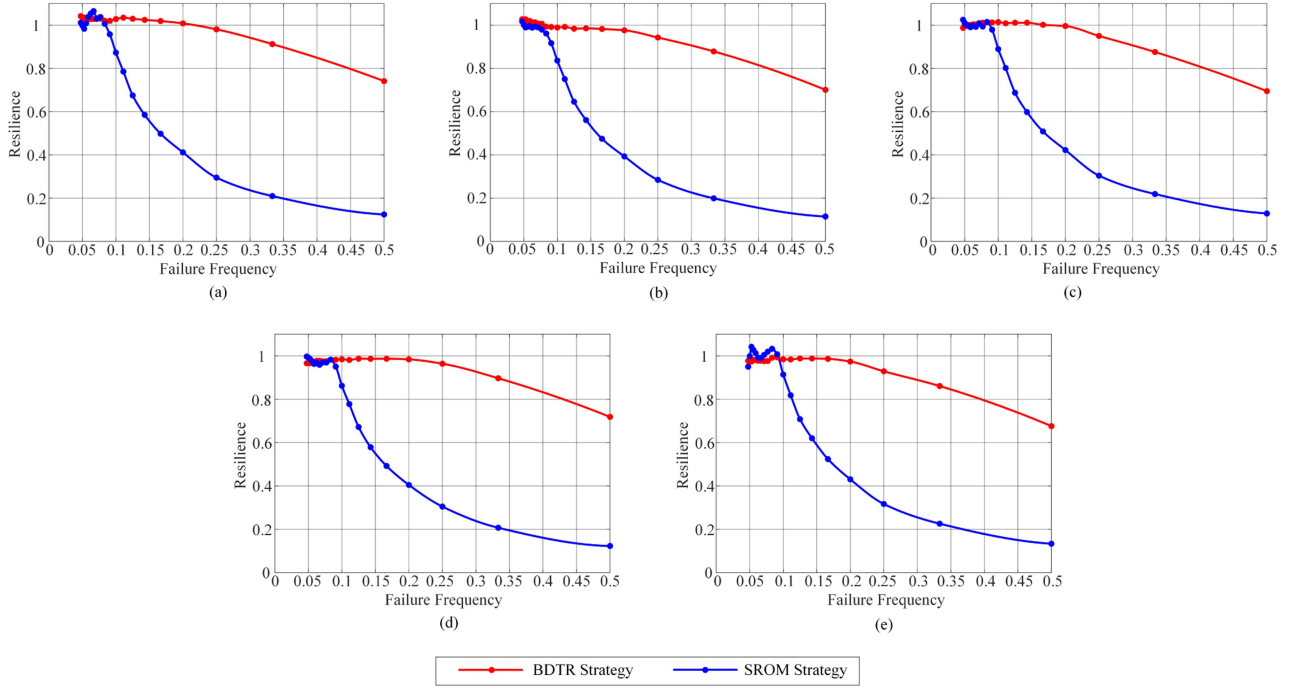


Fig. 9. Resilience comparison of two mission types scenario between BDTR and SROM strategies. (a)–(e) Correspond to the resilience curves of five individual UAVs with the growth of failure frequency. (a) UAV1. (b) UAV2. (c) UAV3. (d) UAV4. (e) UAV5.

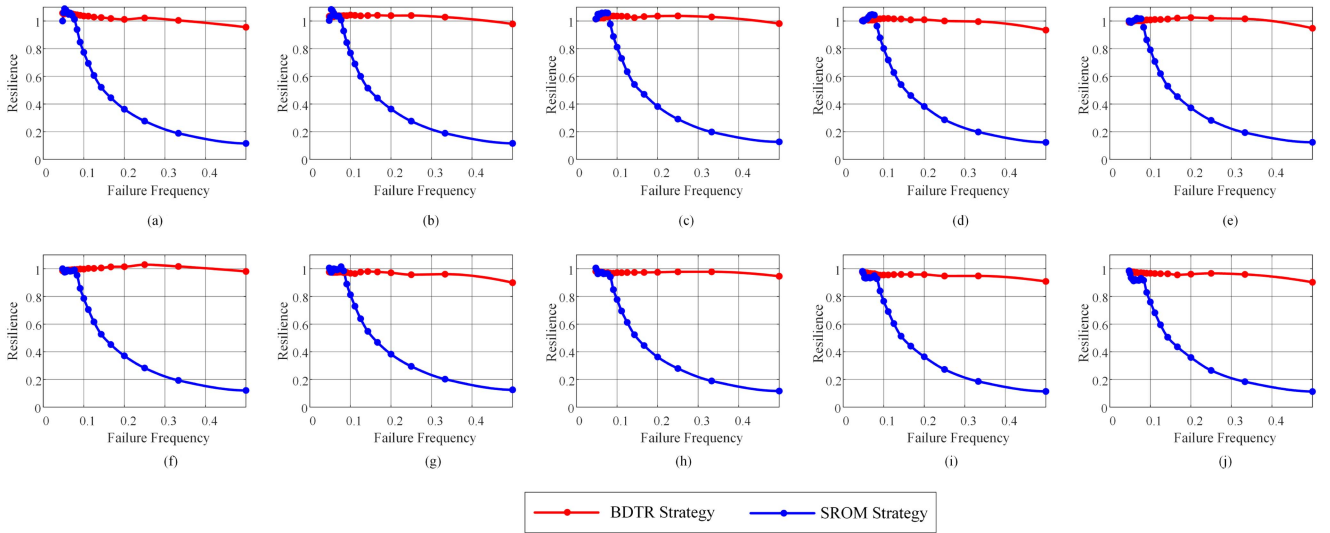


Fig. 10. Resilience comparison of three mission types scenario between BDTR and SROM strategies. (a)–(j) Correspond to the resilience curves of 10 individual UAVs with the growth of failure frequency. (a) UAV1. (b) UAV2. (c) UAV3. (d) UAV4. (e) UAV5. (f) UAV6. (g) UAV7. (h) UAV8. (i) UAV9. (j) UAV10.

was compared against the two baseline strategies, RRAM and SROM, with respect to two key metrics: Mission Completion Rate and the integrated Resilience Index (CRI).

1) Mission Completion Rate Distribution Analyze: The Mission Completion Rate is the primary metric for evaluating whether the swarm can overcome disturbances and achieve its fundamental mission objectives. The simulation results clearly demonstrate the significant differences between the strategies.

As shown in Fig. 11(b), the BDTR strategy achieved a high average mission completion rate of 0.829 across 300 simulations, whereas the baseline SROM strategy only reached 0.225. This comparison strongly demonstrates the necessity of a task reallocation mechanism. The SROM strategy, due to its inability to handle the tasks of failed UAVs, leads to a large number of permanently blocked tasks and thus a very low mission completion rate. Meanwhile, the data in Fig. 11(a) indicates that

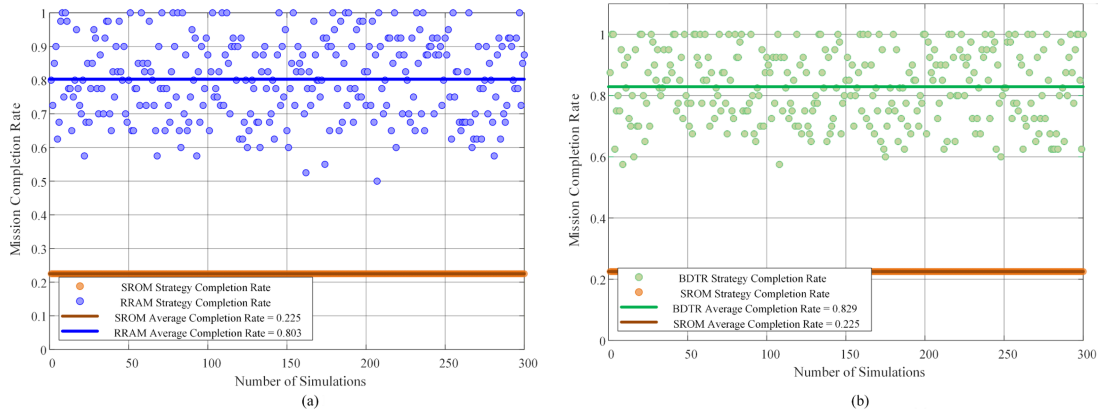


Fig. 11. Comparison of overall mission complete rate distribution under 300 Monte Carlo simulations. (a) Mission complete rate distribution between SROM and RRAM. (b) Mission complete rate distribution between SROM and BDTR.

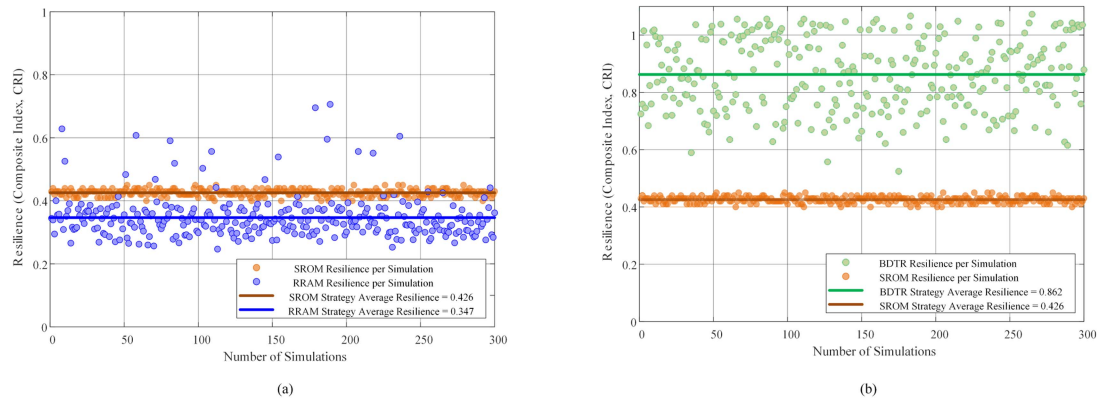


Fig. 12. Comparison of resilience distribution under 300 Monte Carlo simulations. (a) Resilience distribution between SROM and RRAM. (b) Resilience distribution between SROM and BDTR.

the RRAM strategy achieves a completion rate of 0.803, which is comparable to the 0.829 achieved by BDTR. This is because RRAM employs a random selection mechanism that prioritizes task assignment speed over constraint feasibility, allowing it to bypass the blocking issues faced by SROM. However, BDTR still maintains the leading position by dynamically balancing supply and demand.

2) *Resilience Distribution Analyze*: The integrated Resilience Index is a more comprehensive evaluation metric as it also considers the efficiency of mission completion. As shown in Fig. 12(b), the average resilience value of the BDTR strategy reached 0.862, significantly higher than the 0.426 of the SROM strategy, and its resilience values are more concentrated in the high-score range. This indicates that BDTR can not only complete more tasks, but also can do so in a more time-efficient manner.

As illustrated in Fig. 12(a), although the RRAM strategy achieves a mission completion rate comparable to BDTR, its average resilience (0.347) is significantly lower than that of SROM (0.426). This phenomenon is attributed to the dual constraints of the CRI metric regarding “mission output” and “temporal cost.” Specifically, while SROM yields lower output, it avoids substantial temporal penalties through a “fast-fail” mechanism. Conversely, although RRAM enhances throughput, the resulting

severe load imbalance causes specific UAVs to become overloaded, drastically prolonging the total mission duration and consequently diminishing overall resilience. In contrast, BDTR (0.862) effectively resolves this trade-off by maintaining load balance. By maximizing the mission completion rate without compromising execution speed, it achieves a dual optimization of output and efficiency, thereby demonstrating a resilience level significantly superior to the comparative methods.

In summary, the results of the 300 Monte Carlo simulations provide strong evidence that the BDTR strategy is significantly superior to both the RRAM and SROM strategies in terms of both Mission Completion Rate and the integrated Resilience Index. Its core “bidirectional swap” mechanism effectively overcomes the inherent tradeoff between “mission completion rate” and “mission execution efficiency” found in simpler reallocation strategies, making it an excellent solution for enhancing the resilience of UAV swarms in dynamic failure scenarios.

VI. CONCLUSION

To address the dynamic mission capability changes caused by payload failures in UAV swarms operating in hybrid-mission scenarios, this article delivers three key contributions to enhance swarm resilience. First, it constructs a *payload-oriented mission*

throughput framework—this framework breaks through the limitations of traditional binary failure models by directly linking payload-level functional changes to the swarm’s mission execution performance, enabling fine-grained capture of UAV residual capabilities. Second, it proposes a *dynamic resilience assessment model* that comprehensively integrates two core dimensions: mission completion rate and execution efficiency, avoiding one-sided evaluation of either metric alone. Third, it designs and validates the *BDTR algorithm*, which adopts a “failed task offloading and compensatory task acquisition” bidirectional mechanism.

BDTR replenishes failed UAVs with new executable tasks to maximize the swarm’s effective workforce, while alleviating overload on healthy UAVs to maintain system-level load balancing—ultimately achieving simultaneous optimization of mission completion rate and execution efficiency, a critical advantage over conventional one-way reallocation strategies. Simulation results further confirm the effectiveness of these contributions: the payload-oriented framework accurately reflects performance degradation under different disturbance modes, and BDTR outperforms baseline strategies in both mission completion and resilience.

Future research will focus on incorporating these physical constraints into the experimental environment to formulate a multiobjective optimization problem involving communication delay and energy consumption. In addition, exploring multiagent reinforcement learning methods to enable the swarm to autonomously learn optimal reallocation strategies represents a valuable direction to further enhance adaptability in complex dynamic environments.

REFERENCES

- [1] P. Cao et al., “Uav swarm cooperative search based on scalable multi-agent deep reinforcement learning with digital twin-enabled sim-to-real transfer,” *IEEE Trans. Mobile Comput.*, vol. 24, no. 6, pp. 5173–5188, Jun. 2025.
- [2] Y. Hou, J. Zhao, R. Zhang, X. Cheng, and L. Yang, “Uav swarm cooperative target search: A multi-agent reinforcement learning approach,” *IEEE Trans. Intell. Veh.*, vol. 9, no. 1, pp. 568–578, Jan. 2024.
- [3] Y. Wan, Y. Zhong, A. Ma, and L. Zhang, “An accurate uav 3-D path planning method for disaster emergency response based on an improved multiobjective swarm intelligence algorithm,” *IEEE Trans. Cybern.*, vol. 53, no. 4, pp. 2658–2671, Apr. 2023.
- [4] L. Zhang, X. Ma, Z. Zhuang, H. Xu, V. Sharma, and Z. Han, “Q-learning aided intelligent routing with maximum utility in cognitive UAV swarm for emergency communications,” *IEEE Trans. Veh. Technol.*, vol. 72, no. 3, pp. 3707–3723, Mar. 2023.
- [5] L. Kong, L. Wang, Z. Cao, and X. Wang, “Resilience evaluation of uav swarm considering resource supplementation,” *Rel. Eng. Syst. Saf.*, vol. 241, 2024, Art. no. 109673.
- [6] M. Eser and A. E. Yilmaz, “A gossip-based auction algorithm for decentralized task rescheduling in heterogeneous drone swarms,” *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 3, pp. 6673–6696, Jun. 2025.
- [7] J. Li, G. Sun, L. Duan, and Q. Wu, “Multi-objective optimization for uav swarm-assisted iot with virtual antenna arrays,” *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 4890–4907, May 2024.
- [8] X. Zhang, J. Zheng, T. Su, M. Ding, and H. Liu, “An effective dynamic constrained two-archive evolutionary algorithm for cooperative search-track mission planning by UAV swarms in air intelligent transportation,” *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 1, pp. 944–958, Jan. 2024.
- [9] K. Meng, X. He, Q. Wu, and D. Li, “Multi-UAV collaborative sensing and communication: Joint task allocation and power optimization,” *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 4232–4246, Jun. 2023.
- [10] K. Athira et al., “ACO-DTSP algorithm: Optimizing UAV swarm routes with workload constraints,” *Procedia Comput. Sci.*, vol. 235, pp. 163–172, 2024.
- [11] G. Bai, Y. Li, Y. Fang, Y.-A. Zhang, and J. Tao, “Network approach for resilience evaluation of a UAV swarm by considering communication limits,” *Rel. Eng. Syst. Saf.*, vol. 193, 2020, Art. no. 106602.
- [12] Q. Peng, H. Wu, N. Li, and F. Wang, “A dynamic task allocation method for unmanned aerial vehicle swarm based on wolf pack labor division model,” *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 6, pp. 4075–4089, Dec. 2024.
- [13] Q. Wang, D. Zhuang, X. Qu, and H. Xie, “Trajectory prediction of UAV swarm based on neural relational inference model without physical control law,” in *Proc. 39th Chin. Control Conf.*, IEEE, 2020, pp. 7047–7054.
- [14] H. Dui, Y. Lu, and S. Wu, “Competing risks-based resilience approach for multi-state systems under multiple shocks,” *Rel. Eng. Syst. Saf.*, vol. 242, 2024, Art. no. 109773.
- [15] Y. He et al., “A systematic resilience assessment framework for multi-state systems based on physics-informed neural network,” *Rel. Eng. Syst. Saf.*, vol. 257, 2025, Art. no. 110866.
- [16] Z. Sheng, H. Yang, P. Song, S. Yang, and J. Wang, “Towards resilience optimization: Distributed task replanning of multi-UAV system under complex terrain, individual destruction and constrained communication,” *Aerosp. Sci. Technol.*, vol. 168, 2025, Art. no. 110791.
- [17] Z. Zhen, L. Wen, B. Wang, Z. Hu, and D. Zhang, “Improved contract network protocol algorithm based cooperative target allocation of heterogeneous UAV swarm,” *Aerosp. Sci. Technol.*, vol. 119, 2021, Art. no. 107054.
- [18] X. Zheng, F. Zhang, T. Song, and D. Lin, “Heterogeneous multi-UAV distributed task allocation based on CBBA,” in *Proc. IEEE Int. Conf. Unmanned Syst.*, IEEE, 2019, pp. 704–709.
- [19] J. C. Amorim, V. Alves, and E. P. de Freitas, “Assessing a swarm-gap based solution for the task allocation problem in dynamic scenarios,” *Expert Syst. Appl.*, vol. 152, 2020, Art. no. 113437.
- [20] R. Han, T. Zhang, B. Hu, and C. Li, “Distributed auction based-adaptive task assignment and re-assignment for multi-UAV suppressive jamming,” *Defence Technol.*, 2025.
- [21] C. Zhang et al., “A dynamic parameters genetic algorithm for collaborative strike task allocation of unmanned aerial vehicle clusters towards heterogeneous targets,” *Appl. Soft Comput.*, vol. 175, 2025, Art. no. 113075.
- [22] Y. Zeng, L. Wu, J. Li, X. Zhuang, and C. Wu, “Resilient task allocation for UAV swarms: A bilevel PSO-ILP optimization approach,” *Drones*, vol. 9, no. 9, 2025, Art. no. 623.
- [23] T. Zaza and A. Richards, “Ant colony optimization for routing and tasking problems for teams of UAVs,” in *Proc. UKACC Int. Conf. Control*, IEEE, 2014, pp. 652–655.
- [24] C. Lv, Y. Lei, Y. Zhang, D. Duan, and S. Si, “Resilience of the interdependent network against cascade failure,” *Chaos, Solitons Fractals*, vol. 192, 2025, Art. no. 116064.
- [25] J. Li et al., “Network resilience assessment and reinforcement strategy against cascading failure,” *Chaos, Solitons Fractals*, vol. 160, 2022, Art. no. 112271.
- [26] Y. Zhong, H. Li, Q. Sun, Z. Huang, and Y. Zhang, “A kill chain optimization method for improving the resilience of unmanned combat system-of-systems,” *Chaos, Solitons Fractals*, vol. 181, 2024, Art. no. 114685.
- [27] Y. Zhong, H. Li, and X. Zhuang, “A resilience-driven two-stage operational chain optimization model for unmanned weapon system-of-systems under limited resource environments,” *Eksploracja i Niezawodność*, vol. 26, no. 3, 2024.
- [28] Q. Sun, H. Li, Y. Wang, and Y. Zhang, “Multi-swarm-based cooperative reconfiguration model for resilient unmanned weapon system-of-systems,” *Rel. Eng. Syst. Saf.*, vol. 222, 2022, Art. no. 108426.
- [29] H. T. Tran, M. Balchanos, J. C. Domergant, and D. N. Mavris, “A framework for the quantitative assessment of performance-based system resilience,” *Rel. Eng. Syst. Saf.*, vol. 158, pp. 73–84, Feb. 2017.
- [30] K. Moshksar, A. Bayesteh, and A. K. Khandani, “Randomized resource allocation in decentralized wireless networks,” *IEEE Trans. Inf. Theory*, vol. 57, no. 4, pp. 2115–2142, Apr. 2011.
- [31] H. Li, Q. Sun, Y. Zhong, Z. Huang, and Y. Zhang, “A soft resource optimization method for improving the resilience of UAV swarms under continuous attack,” *Rel. Eng. Syst. Saf.*, vol. 237, 2023, Art. no. 109368.
- [32] H. Li, Y. Zhong, and X. Zhuang, “A soft resource optimization method based on autonomous coordination of unmanned swarms system driven by resilience,” *Rel. Eng. Syst. Saf.*, vol. 249, 2024, Art. no. 110227.
- [33] Y. Yang, X. Guo, X. Hai, Q. Feng, B. Sun, and Z. Wang, “Modeling and vulnerability analysis of UAV swarm based on two-layer multi-edge complex network,” *Rel. Eng. Syst. Saf.*, vol. 256, 2025, Art. no. 110779.

- [34] C. Cheng, G. Bai, Y.-A. Zhang, and J. Tao, "Resilience evaluation for UAV swarm performing joint reconnaissance mission," *Chaos: An Interdiscipl. J. Nonlinear Sci.*, vol. 29, no. 5, 2019, Art. no. 053132.
- [35] Z. Wei, Z. Wang, Z. Meng, N. Zhang, H. Wu, and Z. Feng, "Throughput of hybrid UAV networks with scale-free topology," *IEEE Trans. Commun.*, vol. 70, no. 12, pp. 7941–7956, Dec. 2022.



Yifan Zeng was born in 1996. He received the B.E. degree in electrical engineering and automation from Liaoning University, Liaoning, China, in 2018, and the M.S. degree in electronic engineering from the University of Florida, FL, USA, in 2021.

He is currently a Ph.D. candidate with Sun Yat-sen University, Guangzhou, China. His research interests include UAV swarm resilience, task allocation, and AI-driven decision support systems.



Linghua Wu was born in 1994. She received the B.E. degree in applied physics from Inner Mongolia University of Science and Technology, Baotou, China, in 2017, and the M.S. degree in optics from South China Normal University, Guangzhou, China, in 2022.

She is currently a Ph.D. candidate with Sun Yat-sen University, Guangzhou. Her research interests include UAV swarm path planning and resilience.



Jinning Li was born in 1995. He received the B.E. degree in computer science from Liaoning Technical University, Fuxin, China, in 2018. He is currently working toward the Ph.D. degree in electronic information with Sun Yat-sen University, Guangzhou, China.

His research interests include task allocation, intelligent decision-making, and artificial intelligence.



Xuebin Zhuang was born in 1983. He received the B.E. degree in electronic engineering from the Department of Electronic Engineering, Xiamen University, Xiamen, China, in 2006, and the Ph.D. degree in electronic engineering from the Department of Electronic Engineering, Tsinghua University, Beijing, China, in 2011.

He is currently a Professor with Sun Yat-sen University, Guangzhou, China. His research interests include intelligent game confrontation, resilient system design, and resilience assessment.